

HELIOPATH PHASE 4

Smart IoT-Based Solar Pathway Automation System

1. Introduction

HELIOPATH is a sustainable smart campus infrastructure project focused on integrating solar-powered pathways, energy-efficient systems, and smart automation technologies into a modern campus environment. The project combines CAD-based infrastructure modeling with IoT-based automation systems to improve energy management, smart lighting, and campus sustainability.

After completing the earlier infrastructure prototypes, Phase 4 of the project focuses on the implementation of IoT automation using ESP32 microcontrollers, sensors, OLED monitoring systems, and automated pathway lighting logic.

The objective of this phase is to transform HELIOPATH from a static infrastructure model into an intelligent smart campus system capable of environmental sensing, automated control, and energy optimization.

2. Objectives

The main objectives of HELIOPATH Phase 4 are:

- 🚧 To implement smart pathway lighting automation using ESP32
 - 🚧 To monitor environmental light intensity using an LDR sensor
 - 🚧 To control pathway LEDs automatically based on surrounding light conditions
 - 🚧 To display real-time monitoring information using an OLED display
 - 🚧 To simulate smart energy-saving logic for campus infrastructure
 - 🚧 To understand IoT system architecture and embedded automation concepts
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3. Components Used

Component	Purpose
ESP32 Microcontroller	Main processing and control unit
LDR Sensor Module	Detection of ambient light intensity
OLED Display (SSD1306)	Real-time system monitoring
LEDs	Simulated smart pathway lights
Resistors	Current limiting for LEDs
Jumper Wires	Circuit connections
Wokwi Simulator	Virtual IoT simulation platform

4. Software and Tools Used

Software / Tool	Purpose
Wokwi Simulator	ESP32 IoT simulation
Arduino Framework	Embedded programming
FreeCAD	Infrastructure CAD modeling
C++ Programming	ESP32 logic development

5. Methodology

The development of HELIOPATH Phase 4 was completed in multiple stages. Initially, the ESP32 microcontroller environment was studied to understand GPIO functionality, analog input systems, and embedded control logic.

After understanding the basics, an LDR sensor was connected to the ESP32 to measure ambient light intensity. The sensor values were read using analog input pins and displayed through the Serial Monitor for debugging and monitoring purposes.

An OLED display module was then integrated with the ESP32 using I2C communication. The OLED display was used to visualize real-time light intensity values and pathway lighting status.

The lighting automation system was developed using conditional decision-making logic. Different lighting conditions were created based on surrounding light intensity values:

- 🌞 Bright Environment → All LEDs OFF
- 🌤 Medium Light → Partial LEDs ON
- 🌙 Dark Environment → All LEDs ON

This logic simulates intelligent energy-efficient campus pathway lighting behavior.

The entire IoT system was simulated and tested using the Wokwi online ESP32 simulator platform.

6. System Workflow

Step No.	Process	Description
1	Ambient Light Detection	Surrounding light conditions are detected from the environment
2	LDR Sensor Input	The LDR sensor measures the light intensity
3	ESP32 Processing	ESP32 reads the analog sensor values
4	Decision Logic Execution	The programmed conditions compare light intensity levels
5	Automatic LED Control	Pathway LEDs are automatically switched ON/OFF based on light conditions
6	OLED Monitoring Output	System status and light intensity values are displayed on the OLED screen

7. Implemented Features

The following features were successfully implemented during Phase 4:

- 🔧 Smart pathway lighting automation
 - 🔧 Multi-level LED lighting system
 - 🔧 Real-time ambient light sensing
 - 🔧 OLED-based monitoring interface
 - 🔧 Automatic energy-saving logic
 - 🔧 Real-time Serial Monitor debugging
 - 🔧 Sensor-based infrastructure control
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8. Lighting Control Logic

Light	Intensity	Condition	LED Behavior
Bright	Light	All LEDs	OFF
Medium	Light	Half LEDs	ON
Low Light	Darkness	All LEDs	ON

This logic helps simulate smart energy conservation within the HELIOPATH campus environment.

9. Results and Observations

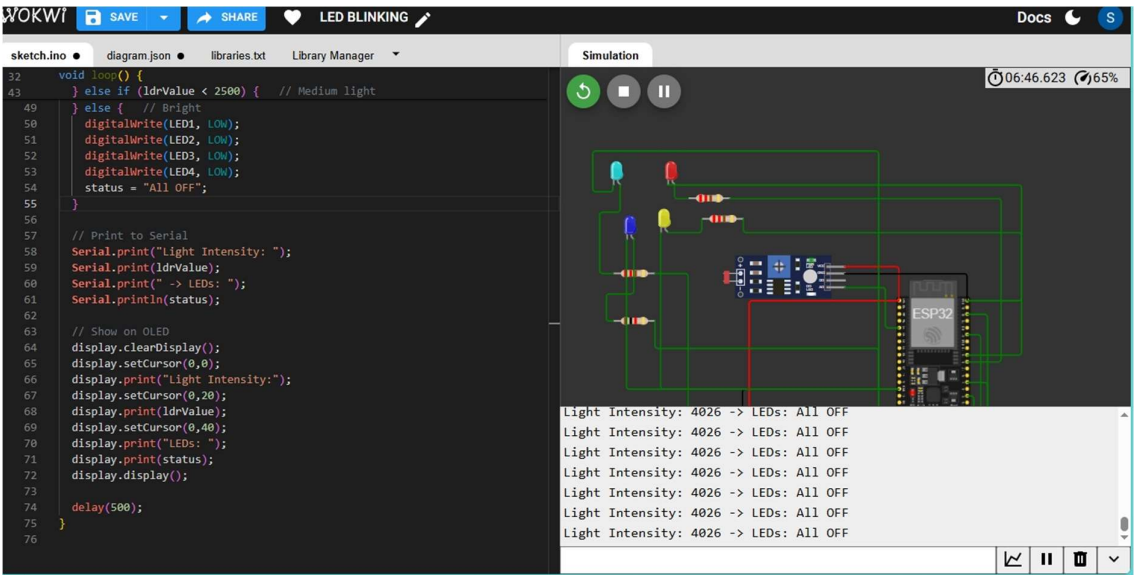
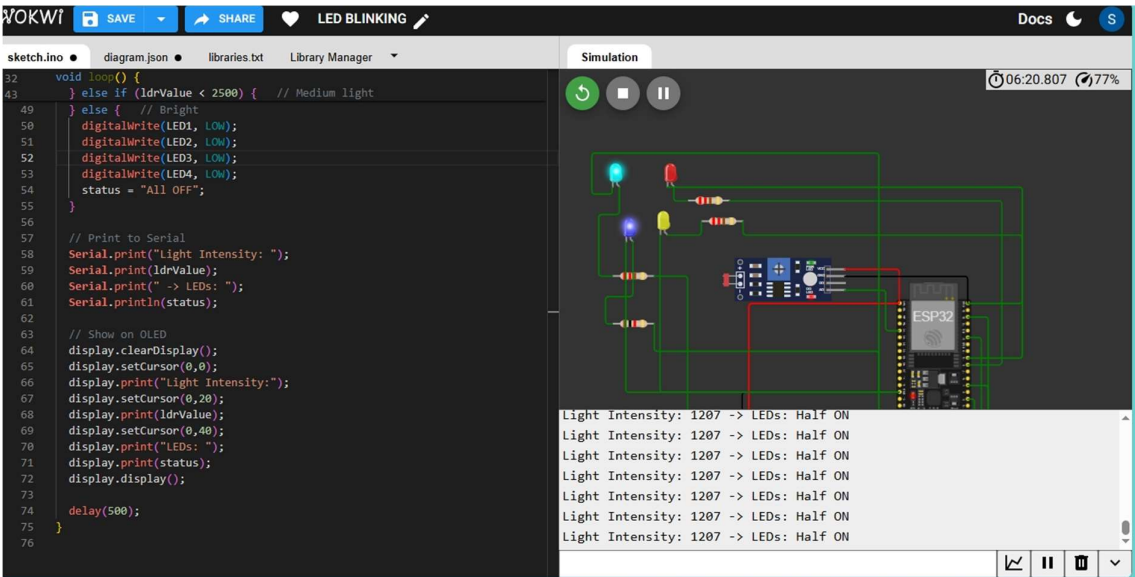
The implemented system successfully demonstrated automatic lighting behavior based on environmental conditions.

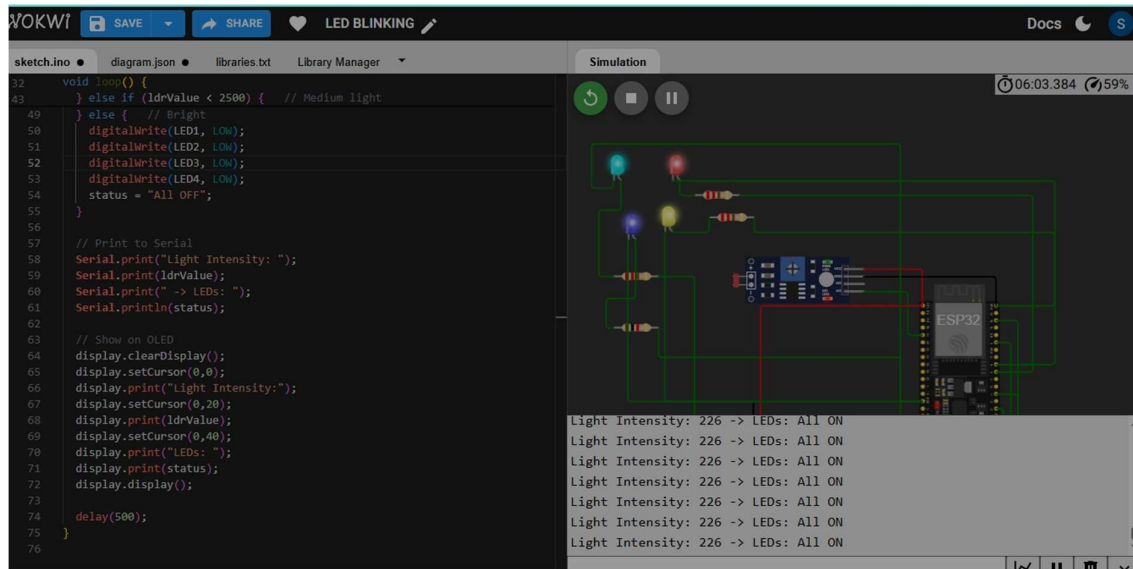
The LDR sensor continuously monitored surrounding light intensity and transmitted analog data to the ESP32. Based on predefined threshold conditions, the ESP32 controlled the LEDs automatically.

The OLED display successfully displayed:

- Real-time light intensity values
- LED operational status
- Lighting mode conditions

The project demonstrated how IoT systems can improve energy efficiency and automation in smart campus infrastructure.





10. Learning Outcomes

During the development of HELIOPATH Phase 4, the following concepts were learned:

- ✚ ESP32 microcontroller fundamentals
- ✚ GPIO pin configuration
- ✚ Analog sensor reading
- ✚ OLED display interfacing
- ✚ Embedded system programming
- ✚ Conditional automation logic
- ✚ IoT simulation and testing
- ✚ Smart infrastructure system design

11. Future Improvements

The following future upgrades are planned for HELIOPATH:

- ✚ PIR motion sensor integration
- ✚ WiFi-based remote monitoring dashboard

- 🔌 Smart EV charging management
 - 🔌 Battery health monitoring system
 - 🔌 Real-time solar energy analytics
 - 🔌 Cloud-connected campus monitoring
 - 🔌 Mobile application integration
 - 🔌 Intelligent power distribution system
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12. Conclusion

HELIOPATH Phase 4 successfully transformed the project from a static infrastructure model into an intelligent IoT-enabled smart campus system.

The implementation of ESP32-based automation, smart pathway lighting, sensor monitoring, and OLED-based status visualization demonstrated the practical integration of IoT technologies into sustainable infrastructure systems.

This phase also helped in developing practical understanding of embedded systems, automation logic, smart energy management, and real-time monitoring technologies.

The project provides a strong foundation for future development in smart campus infrastructure, renewable energy systems, and intelligent automation technologies.

13. References

- 1) [ESP32 Official Documentation](#)
- 2) [Wokwi ESP32 Simulator](#)
- 3) [Arduino Documentation](#)
- 4) [Adafruit SSD1306 Library Documentation](#)
- 5) [FreeCAD Official Website](#)